

AI Integration in Decision-Support Systems for Insurance Fraud Detection

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Abstract

This paper extends a prior proposal to incorporate artificial intelligence (AI) into the operations of an insurance company by developing and analyzing a decision-support framework for fraud detection. Using a simulated dataset representative of real insurance claims, the study applies three machine learning models—Decision Tree, Random Forest, and Multilayer Perceptron (MLP)—to evaluate the effectiveness of predictive analytics in identifying fraudulent activity. Evaluation metrics such as ROC-AUC and Precision-Recall (PR) curves are used to assess model performance and reliability. The study highlights interpretability, calibration, and human oversight as critical factors in adopting AI for operational decision-support systems. Finally, the discussion explores ethical and governance considerations to ensure responsible and trustworthy deployment.

Keywords: AI, Machine Learning, Fraud Detection, Decision-Support Systems, Random Forest, Ethics, Explainability

1. Introduction

Insurance fraud poses a substantial financial burden on insurers and policyholders worldwide. Traditional rule-based systems often struggle to adapt to new patterns of fraudulent behavior, leading to inefficiencies and losses. With advances in AI and machine learning, insurers now can automate fraud detection processes while preserving human oversight and accountability.

The current research extends that proposal by developing a technical, data-driven framework that integrates AI into the company's decision-support system. The study

focuses on building interpretable models that balance automation with transparency, enabling analysts to make better, faster, and more data-informed decisions.

The objectives of this research are threefold:

- To evaluate the performance of common AI models on an imbalanced fraud dataset.
- To analyze trade-offs between accuracy, interpretability, and operational scalability.
- To demonstrate how predictive models can be embedded into a decision-support workflow that assists but does not replace human experts.

2. Literature Review

Artificial intelligence in the insurance sector has rapidly evolved from experimental pilots to core business functions. PwC (2023) identified that insurers leveraging predictive analytics can reduce claim-processing time by 40% and improve fraud detection accuracy by more than 25%. Similarly, the Cloud Security Alliance (2023) emphasized the importance of secure data governance when deploying AI systems, warning that breaches in model management can lead to reputational and financial damage.

Doshi-Velez and Kim (2017) advocate for interpretable machine learning as a prerequisite for responsible deployment, arguing that decision-support systems must remain explainable to stakeholders. Ribeiro, Singh, and Guestrin (2016) introduced LIME to explain individual model predictions, reinforcing the need for transparency. Pedregosa et al. (2011) demonstrated the use of `scikit-learn` pipelines to ensure reproducibility and consistency in applied machine learning workflows.

Ashkenas and Matta (2021) describe a framework for scaling successful pilot projects, emphasizing small-scale experimentation followed by progressive rollout. Sjödin et al. (2021) add that AI-driven innovation depends not only on algorithms but also on organizational culture, data maturity, and employee training. Together, these studies support a hybrid model of AI adoption that integrates technical innovation with human oversight, transparency, and ethical governance.

3. Methods

3.1 Data Simulation

A synthetic dataset of 25,000 insurance claims was generated to model real-world data. Each record includes features such as policy age, claim amount, prior claims, provider tenure, claim type, and region. A probabilistic mechanism simulates fraud likelihood, ensuring an approximately 8% fraud rate consistent with industry statistics.

3.2 Preprocessing

Data preprocessing used a unified `ColumnTransformer` to standardize numerical variables and one-hot encode categorical features. This ensures that all models receive consistent and comparable inputs. Feature scaling mitigates the impact of extreme values.

3.3 Model Selection

Three supervised learning algorithms were implemented:

- **Decision Tree (CART):** Provides interpretability and visual transparency.
- **Random Forest:** Improves generalization through ensemble averaging.
- **Multilayer Perceptron (MLP):** Models non-linear relationships within data.

All models were trained on 80% of the dataset, with 20% reserved for testing using stratified sampling.

3.4 Evaluation Metrics

Performance was assessed using ROC-AUC and PR-AUC. ROC-AUC captures overall discriminatory ability, while PR-AUC focuses on rare-event classification such as fraud detection. Confusion matrices were analyzed to assess false positives and false negatives.

4. Results

Among the models tested, Random Forest achieved the highest ROC-AUC and PR-AUC scores. Decision Tree was less accurate but highly interpretable. MLP delivered strong performance but required careful tuning.

Model	ROC-AUC	PR-AUC	Notes
Random Forest	0.93	0.74	Best overall performance
Decision Tree	0.86	0.61	Highly interpretable
MLP	0.91	0.71	Complex but effective

Threshold tuning demonstrated operational trade-offs between precision and recall, supporting human-in-the-loop decision-making.

5. Discussion

The findings validate the feasibility of AI-driven fraud detection as part of a decision-support system rather than a fully autonomous solution. Random Forest models, though less transparent, can be augmented with explainability tools such as SHAP or LIME. Decision Trees provide visual interpretability for stakeholders. Integrating AI requires collaboration between data science, IT, and claims departments. Analysts should interpret AI outputs responsibly, understanding predictive probabilities rather than certainties.

Continuous monitoring and retraining are essential to adapt to changes in fraud patterns, regulation, and behavior. Interpretability and auditability sustain trust in AI systems.

6. Ethical and Governance Considerations

Ethical AI in insurance must balance automation with fairness, privacy, and accountability. The Cloud Security Alliance (2023) outlines frameworks for data security, including encryption, access control, and audit mechanisms. Fairness audits and explainability documentation should accompany deployments. Human review ensures appeal mechanisms

and regulatory compliance.

7. Conclusion and Future Work

This research demonstrates a scalable, interpretable, and ethical approach to integrating AI into insurance decision-support systems. Random Forest models provide accuracy; Decision Trees enhance interpretability. Together, they form a hybrid framework augmenting human expertise.

Future work includes incorporating cost-sensitive learning, temporal data modeling (e.g., LSTMs or Transformers), and integrating explainability tools into operational dashboards. Developing secure, cloud-based APIs with monitoring frameworks such as MLflow or EvidentlyAI will further improve scalability and compliance.

GitHub Repository

Repository: <https://github.com/siankop22/insurance-fraud-detection-cs>

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